```
#Change epochs
import numpy as np
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])
epochs = 5
w,b = np.zeros(2),0

for _ in range(epochs):
  for x,y in zip(x_list,y_list):
    z = np.dot(x,w) + b
    y_pred = 1 if z >= 0 else 0
    E = y - y_pred
    print("error : ",E,"<- at epoch",_)
    w += E*x
    b += E
    print(w,b)
```

```
error :  -1 <- at epoch 0
[0. 0.] -1
error :  0 <- at epoch 0
[0. 0.] -1
error :  0 <- at epoch 0
[0. 0.] -1
error :  1 <- at epoch 0
[1. 1.] 0
error :  -1 <- at epoch 1
[1. 1.] -1
error :  -1 <- at epoch 1
[1. 0.] -2
error :  0 <- at epoch 1
[1. 0.] -2
error :  1 <- at epoch 1
[2. 1.] -1
error :  0 <- at epoch 2
[2. 1.] -1
error :  -1 <- at epoch 2
[2. 0.] -2
error :  -1 <- at epoch 2
[1. 0.] -3
error :  1 <- at epoch 2
[2. 1.] -2
error :  0 <- at epoch 3
[2. 1.] -2
error :  0 <- at epoch 3
[2. 1.] -2
error :  -1 <- at epoch 3
[1. 1.] -3
error :  1 <- at epoch 3
[2. 2.] -2
error :  0 <- at epoch 4
[2. 2.] -2
error :  -1 <- at epoch 4
[2. 1.] -3
error :  0 <- at epoch 4
[2. 1.] -3
error :  0 <- at epoch 4
[2. 1.] -3
```

```
#Change random weights and bias
import numpy as np
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])
epochs = 5
w = np.random.randint(0,2,2)
b = np.random.randint(2)

for _ in range(epochs):
  for x,y in zip(x_list,y_list):
    z = np.dot(x,w) + b
    y_pred = 1 if z >= 0 else 0
    E = y - y_pred
    print("error : ",E,"<- at epoch",_)
    w += E*x
    b += E
    print(w,b)
```

```
error :  -1 <- at epoch 0
[1 0] 0
error :  -1 <- at epoch 0
[ 1 -1] -1
error :  -1 <- at epoch 0
[ 0 -1] -2
error :  1 <- at epoch 0
[1 0] -1
error :  0 <- at epoch 1
[1 0] -1
error :  0 <- at epoch 1
[1 0] -1
error :  -1 <- at epoch 1
[0 0] -2
error :  1 <- at epoch 1
[1 1] -1
error :  0 <- at epoch 2
[1 1] -1
error :  -1 <- at epoch 2
[1 0] -2
error :  0 <- at epoch 2
[1 0] -2
error :  1 <- at epoch 2
[2 1] -1
error :  0 <- at epoch 3
[2 1] -1
error :  -1 <- at epoch 3
[2 0] -2
error :  -1 <- at epoch 3
[1 0] -3
error :  1 <- at epoch 3
[2 1] -2
error :  0 <- at epoch 4
[2 1] -2
error :  0 <- at epoch 4
[2 1] -2
error :  -1 <- at epoch 4
[1 1] -3
error :  1 <- at epoch 4
[2 2] -2
```

```
#Add logic to stop the training when error is 'zero' for all the inputs
import numpy as np
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])
epochs = 5
w,b = np.zeros(2),0

for _ in range(epochs):
    all_zero_error = True
    for x,y in zip(x_list,y_list):
        z = np.dot(x,w) + b
        y_pred = 1 if z >= 0 else 0
        E = y - y_pred
        print("error : ",E,"<- at epoch",_)
        if E !=0:
            all_zero_error = False
        w += E*x
        b += E
    if all_zero_error:
        print("Training stop: zero error for all inputs")
        break
```

```
error :  -1 <- at epoch 0
error :  0 <- at epoch 0
error :  0 <- at epoch 0
error :  1 <- at epoch 0
error :  -1 <- at epoch 1
error :  -1 <- at epoch 1
error :  0 <- at epoch 1
error :  1 <- at epoch 1
error :  0 <- at epoch 2
error :  -1 <- at epoch 2
error :  -1 <- at epoch 2
error :  1 <- at epoch 2
error :  0 <- at epoch 3
error :  0 <- at epoch 3
error :  -1 <- at epoch 3
error :  1 <- at epoch 3
error :  0 <- at epoch 4
error :  -1 <- at epoch 4
error :  0 <- at epoch 4
```

```
error :  0 <- at epoch 4
```

```
#OR gate
import numpy as np
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,1,1,1])
epochs = 4
w,b = np.zeros(2),0

for _ in range(epochs):
  for x,y in zip(x_list,y_list):
    z = np.dot(x,w) + b
    y_pred = 1 if z >= 0 else 0
    E = y - y_pred
    print("error : ",E,"<- at epoch",_)
    w += E*x
    b += E
    print(w,b)
```

```
error :  -1 <- at epoch 0
[0. 0.] -1
error :  1 <- at epoch 0
[0. 1.] 0
error :  0 <- at epoch 0
[0. 1.] 0
error :  0 <- at epoch 0
[0. 1.] 0
error :  -1 <- at epoch 1
[0. 1.] -1
error :  0 <- at epoch 1
[0. 1.] -1
error :  1 <- at epoch 1
[1. 1.] 0
error :  0 <- at epoch 1
[1. 1.] 0
error :  -1 <- at epoch 2
[1. 1.] -1
error :  0 <- at epoch 2
[1. 1.] -1
error :  0 <- at epoch 2
[1. 1.] -1
error :  0 <- at epoch 2
[1. 1.] -1
error :  0 <- at epoch 3
[1. 1.] -1
error :  0 <- at epoch 3
[1. 1.] -1
error :  0 <- at epoch 3
[1. 1.] -1
error :  0 <- at epoch 3
[1. 1.] -1
```

```
#XOR gate
import numpy as np
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,1,1,0])
epochs = 5
w,b = np.zeros(2),0

for _ in range(epochs):
  for x,y in zip(x_list,y_list):
    z = np.dot(x,w) + b

    y_pred = 1 if z >= 0 else 0
    E = y - y_pred
    print("error : ",E,"<- at epoch",_)
    w += E*x
    b += E
    print(w,b)
```

```
error :  -1 <- at epoch 0
[0. 0.] -1
error :  1 <- at epoch 0
[0. 1.] 0
error :  0 <- at epoch 0
[0. 1.] 0
error :  -1 <- at epoch 0
[-1.  0.] -1
```

```
error :  0 <- at epoch 1
[-1.  0.] -1
error :  1 <- at epoch 1
[-1.  1.] 0
error :  1 <- at epoch 1
[0. 1.] 1
error :  -1 <- at epoch 1
[-1.  0.] 0
error :  -1 <- at epoch 2
[-1.  0.] -1
error :  1 <- at epoch 2
[-1.  1.] 0
error :  1 <- at epoch 2
[0. 1.] 1
error :  -1 <- at epoch 2
[-1.  0.] 0
error :  -1 <- at epoch 3
[-1.  0.] -1
error :  1 <- at epoch 3
[-1.  1.] 0
error :  1 <- at epoch 3
[0. 1.] 1
error :  -1 <- at epoch 3
[-1.  0.] 0
error :  -1 <- at epoch 4
[-1.  0.] -1
error :  1 <- at epoch 4
[-1.  1.] 0
error :  1 <- at epoch 4
[0. 1.] 1
error :  -1 <- at epoch 4
[-1.  0.] 0
```